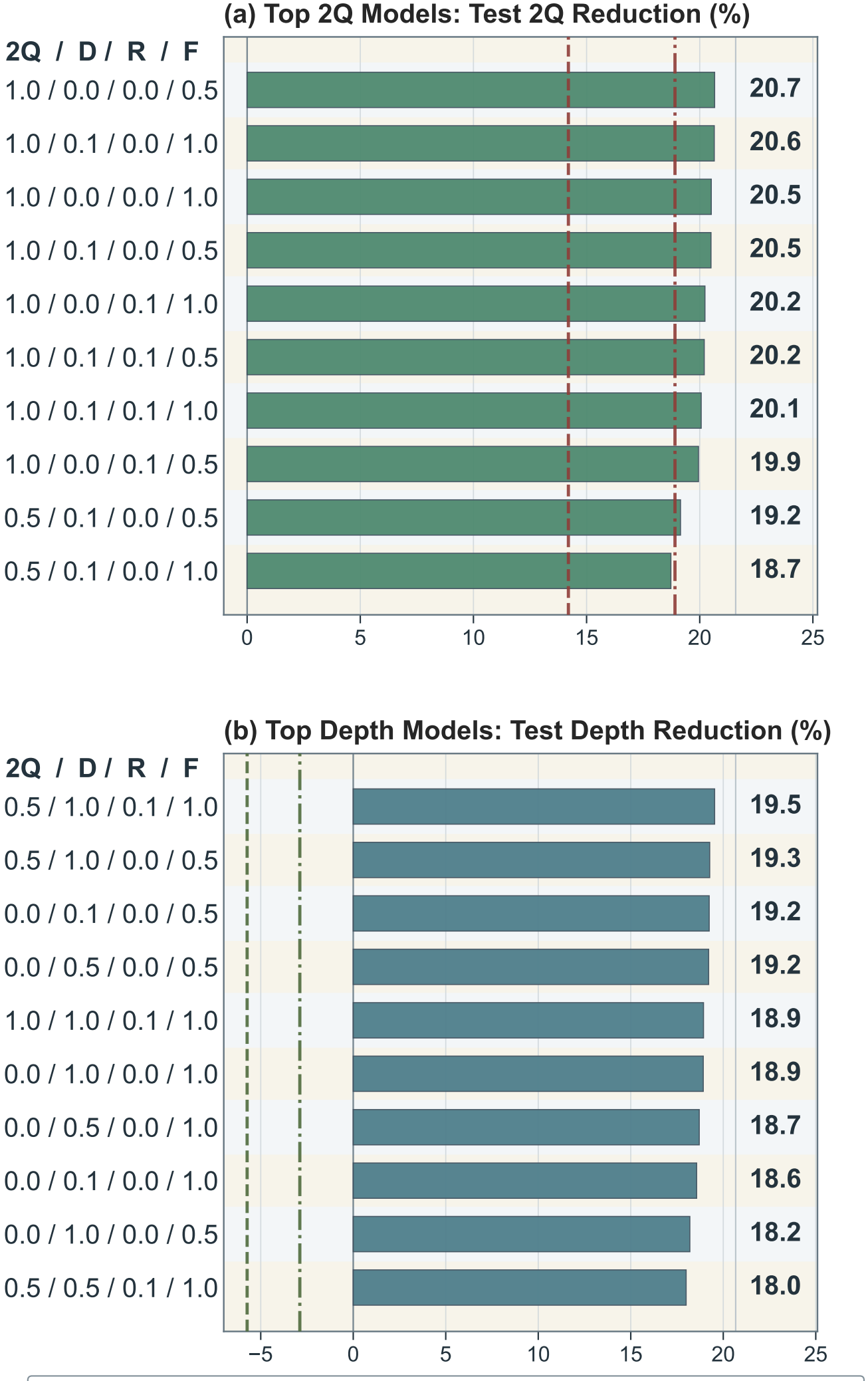Top Reward Configurations by 2Q and Depth Reduction